



Name: _____

Date: _____

Score: / 10

Learning Objectives

- Build and interpret truth tables for $\wedge$, $\vee$, $\neg$, $\rightarrow$, and $\leftrightarrow$
- Apply De Morgan's Laws and logical equivalences
- Evaluate compound propositions and identify tautologies
- Construct and validate logical arguments (Modus Ponens, contrapositive)

For truth tables: always list all 2^n rows for n variables. Use T and F consistently — not 1/0 or True/False.

1. Complete the truth table for $p \wedge q$ (AND) and $p \vee q$ (OR).

p	q	$p \wedge q$	$p \vee q$
T	T	_____	_____
T	F	_____	_____
F	T	_____	_____
F	F	_____	_____

Answer: _____

2. Complete the truth table for $\neg p$, $p \rightarrow q$ (IF p THEN q), and $\neg p \vee q$.

p	q	$\neg p$	$p \rightarrow q$	$\neg p \vee q$
T	T	_____	_____	_____
T	F	_____	_____	_____
F	T	_____	_____	_____
F	F	_____	_____	_____

Answer: _____

3. Complete the truth table to verify De Morgan's Law: $\neg(p \wedge q) \equiv \neg p \vee \neg q$.

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p \vee \neg q$
T	T	_____	_____	_____
T	F	_____	_____	_____
F	T	_____	_____	_____
F	F	_____	_____	_____

Answer: _____



4. Complete the truth table for $p \wedge (q \vee r)$.

p	q	r	$q \vee r$	$p \wedge (q \vee r)$
T	T	T	—	—
T	T	F	—	—
T	F	T	—	—
T	F	F	—	—
F	T	T	—	—
F	T	F	—	—
F	F	T	—	—
F	F	F	—	—

Answer: _____

5. Build a truth table for $(p \rightarrow q) \wedge (q \rightarrow p)$. Is this a tautology, contradiction, or neither?

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	—	—	—
T	F	—	—	—
F	T	—	—	—
F	F	—	—	—

Answer: _____

6. Evaluate the logical argument: "If it rains, the ground is wet. It is raining. Therefore, the ground is wet." Is this valid?

$$p \rightarrow q, \quad p \vdash q$$

Answer: _____

7. Write the converse, inverse, and contrapositive of: "If it snows, then school is cancelled."

$$p \rightarrow q$$

Answer: _____

8. Determine the truth value of $(p \vee \neg q) \rightarrow (\neg p \wedge q)$ when $p = T$ and $q = F$.

$$p = T, \quad q = F \Rightarrow (p \vee \neg q) \rightarrow (\neg p \wedge q)$$

Answer: _____



9. Complete the truth table for $p \leftrightarrow q$ (biconditional, "p if and only if q").

p	q	$p \leftrightarrow q$	$p \rightarrow q$	$q \rightarrow p$
T	T	—	—	—
T	F	—	—	—
F	T	—	—	—
F	F	—	—	—

Answer: _____

10. Use a truth table to show whether $p \rightarrow q$ and $\neg q \rightarrow \neg p$ are logically equivalent.

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	—	—	—	—
T	F	—	—	—	—
F	T	—	—	—	—
F	F	—	—	—	—

Answer: _____





Problems 1–5 are full truth tables — key exam skill. Problem 3 verifies De Morgan's Law by table. Problem 7 introduces the contrapositive (only equivalent transform).

Solutions

1. Complete the truth table for $p \wedge q$ (AND) and $p \vee q$ (OR).

p	q	$p \wedge q$	$p \vee q$
T	T	—	—
T	F	—	—
F	T	—	—
F	F	—	—

- $p \wedge q$ (AND) is T only when BOTH p and q are T.
- $p \vee q$ (OR) is T when AT LEAST ONE of p, q is T.
- Row 1: $T \wedge T = T$, $T \vee T = T$. Row 2: $T \wedge F = F$, $T \vee F = T$.
- Row 3: $F \wedge T = F$, $F \vee T = T$. Row 4: $F \wedge F = F$, $F \vee F = F$.

Answer:

p	q	$p \wedge q$	$p \vee q$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	F

2. Complete the truth table for $\neg p$, $p \rightarrow q$ (IF p THEN q), and $\neg p \vee q$.

p	q	$\neg p$	$p \rightarrow q$	$\neg p \vee q$
T	T	—	—	—
T	F	—	—	—
F	T	—	—	—
F	F	—	—	—

- $\neg p$ flips the truth value of p.
- $p \rightarrow q$ (conditional) is F only when p is T and q is F.
- Key insight: $p \rightarrow q$ is logically equivalent to $\neg p \vee q$ (same column!).
- This equivalence is used to prove theorems by contrapositive.

Answer:

p	q	$\neg p$	$p \rightarrow q$	$\neg p \vee q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T



3. Complete the truth table to verify De Morgan's Law: $\neg(p \wedge q) \equiv \neg p \vee \neg q$.

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p \vee \neg q$
T	T	—	—	—
T	F	—	—	—
F	T	—	—	—
F	F	—	—	—

→ Columns 4 and 5 are identical → the two expressions are logically equivalent.

→ De Morgan's Law 1: $\neg(p \wedge q) \equiv \neg p \vee \neg q$.

→ De Morgan's Law 2: $\neg(p \vee q) \equiv \neg p \wedge \neg q$.

→ These laws let you distribute negation across AND/OR.

Answer:

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p \vee \neg q$
T	T	T	F	F
T	F	F	T	T
F	T	F	T	T
F	F	F	T	T

4. Complete the truth table for $p \wedge (q \vee r)$.

p	q	r	$q \vee r$	$p \wedge (q \vee r)$
T	T	T	—	—
T	T	F	—	—
T	F	T	—	—
T	F	F	—	—
F	T	T	—	—
F	T	F	—	—
F	F	T	—	—
F	F	F	—	—

→ First evaluate $q \vee r$ (OR) for each row.

→ Then evaluate $p \wedge (q \vee r)$ — AND with result from column 4.

→ $p \wedge (q \vee r)$ is T only when p is T AND at least one of q, r is T.

→ Rows 1–3 are T; row 4 (T,F,F) gives F because $q \vee r = F$.

Answer:

p	q	r	$q \vee r$	$p \wedge (q \vee r)$
T	T	T	T	T
T	T	F	T	T
T	F	T	T	T
T	F	F	F	F
F	T	T	T	F
F	T	F	T	F
F	F	T	T	F
F	F	F	F	F



5. Build a truth table for $(p \rightarrow q) \wedge (q \rightarrow p)$. Is this a tautology, contradiction, or neither?

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	—	—	—
T	F	—	—	—
F	T	—	—	—
F	F	—	—	—

→ $(p \rightarrow q) \wedge (q \rightarrow p) = T$ only in rows 1 and 4 (when p and q have the same value).

→ This is the biconditional $p \leftrightarrow q$ — neither always T (tautology) nor always F .

→ It is a contingency (sometimes T , sometimes F).

Answer:

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

6. Evaluate the logical argument: "If it rains, the ground is wet. It is raining. Therefore, the ground is wet." Is this valid?

$p \rightarrow q, p \vdash q$

→ Let $p =$ "It is raining", $q =$ "The ground is wet".

→ Premise 1: $p \rightarrow q$. Premise 2: p .

→ Modus Ponens rule: from $p \rightarrow q$ and p , we can conclude q .

→ The argument form is valid (always produces T when premises are T).

Answer: Valid — Modus Ponens

7. Write the converse, inverse, and contrapositive of: "If it snows, then school is cancelled."

$p \rightarrow q$

→ Original ($p \rightarrow q$): "If it snows, school is cancelled."

→ Converse ($q \rightarrow p$): "If school is cancelled, then it snowed." (NOT equivalent)

→ Inverse ($\neg p \rightarrow \neg q$): "If it does not snow, school is not cancelled." (NOT equivalent)

→ Contrapositive ($\neg q \rightarrow \neg p$): "If school is not cancelled, it did not snow." (EQUIVALENT)

→ Only the contrapositive is logically equivalent to the original.

Answer: Contrapositive: $\neg q \rightarrow \neg p \equiv p \rightarrow q$

8. Determine the truth value of $(p \vee \neg q) \rightarrow (\neg p \wedge q)$ when $p = T$ and $q = F$.

$p = T, q = F \Rightarrow (p \vee \neg q) \rightarrow (\neg p \wedge q)$

→ $\neg q = \neg F = T$. $p \vee \neg q = T \vee T = T$.

→ $\neg p = \neg T = F$. $\neg p \wedge q = F \wedge F = F$.

→ $(p \vee \neg q) \rightarrow (\neg p \wedge q) = T \rightarrow F = F$.

→ A conditional is F only when the hypothesis is T and conclusion is F .

Answer: $T \rightarrow F = F$



9. Complete the truth table for $p \leftrightarrow q$ (biconditional, "p if and only if q").

p	q	$p \leftrightarrow q$	$p \rightarrow q$	$q \rightarrow p$
T	T	—	—	—
T	F	—	—	—
F	T	—	—	—
F	F	—	—	—

→ $p \leftrightarrow q$ is T when p and q have the SAME truth value.

→ Equivalently: $p \leftrightarrow q = (p \rightarrow q) \wedge (q \rightarrow p)$.

→ Notice: $p \leftrightarrow q$ column matches rows where $p=q$.

Answer:

p	q	$p \leftrightarrow q$	$p \rightarrow q$	$q \rightarrow p$
T	T	T	T	T
T	F	F	F	T
F	T	F	T	F
F	F	T	T	T

10. Use a truth table to show whether $p \rightarrow q$ and $\neg q \rightarrow \neg p$ are logically equivalent.

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	—	—	—	—
T	F	—	—	—	—
F	T	—	—	—	—
F	F	—	—	—	—

→ Columns 5 and 6 are identical → $p \rightarrow q \equiv \neg q \rightarrow \neg p$.

→ $\neg q \rightarrow \neg p$ is the CONTRAPOSITIVE of $p \rightarrow q$.

→ A conditional and its contrapositive are always logically equivalent.

Answer:

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

